COMP20008 Assignment 2

Which Flower Measurements Distinguish Iris Species? Prediction, Feature Relevance and Natural Groupings

SAMPLE REPORT prepared by the teaching team for illustration only. The structure follows the A2 specification for a group of four.

Group name	W00G00
Group members	Jon Snow (1000001) Arya Stark (1000002) Tyrion Lannister (1000003) Danny Targaryen (1000004)

1. Introduction

Research question. *Can an iris flower's species be predicted from its four sepal and petal measurements, and which measurements distinguish the three species most strongly?*

Identifying plant species from simple physical measurements is a classic task in taxonomy, and understanding which measurements carry the most information can make identification faster and cheaper. We use Fisher's (1936) Iris dataset as distributed in scikit-learn (Pedregosa et al., 2011). It contains 150 flowers from three species (setosa, versicolor, virginica; 50 each), each described by sepal length, sepal width, petal length and petal width in centimetres. We added an *id* column (1–150) in the original row order so that individual flowers can be referenced. The analysis follows the RQ through preprocessing, correlation analysis, supervised learning (KNN and Decision Tree), feature selection, and PCA and clustering.

2. Methodology

2.1. Data Preprocessing

We considered six candidate preprocessing steps (Table 1) and implemented three. All values in Table 1 were computed on the full dataset before any step was applied.

Table 1: Candidate preprocessing steps, the dataset property that informed each decision, and the outcome.

#	Candidate step	Dataset property (computed)	Decision
1	Remove exact duplicate rows	1 duplicate pair (ids 102 and 143, both virginica); 0.7% of rows	Selected
2	Engineer petal_area = petal_length × petal_width	petal_length and petal_width Pearson r = 0.963	Selected
3	Z-score standardisation (fitted on training data only)	Standard deviations range from 0.436 (sepal_width) to 1.768 (petal_length)	Selected
4	IQR outlier removal	4 rows (2.7%) flagged, all on sepal_width (ids 16, 33, 34, 61)	Not selected
5	Min–max scaling to [0, 1]	sepal_width range 2.0–4.4 cm is set by the 4 IQR outliers	Not selected
6	Discretise petal_length into 3 equal-width bins	8 of 149 flowers (5.4%) fall in a bin dominated by another species	Not selected

Why these three. Duplicate removal was selected because a duplicated flower (ids 102 and 143) could appear in both a training fold and a validation fold, which would inflate cross-validation scores. petal_area was selected because petal_length and petal_width are very strongly correlated (r = 0.963), so a single combined feature summarises overall petal size. Standardisation was selected because KNN and K-Means use Euclidean distance, and the features have different spreads (0.436 to 1.768), so without scaling the features with larger values would dominate the distance calculation.

Why not the other three. IQR outlier removal was rejected because the 4 flagged rows (2.7%) are genuine measurements rather than recording errors, and removing them would reduce an already small dataset. Min–max scaling was rejected because it is sensitive to extreme values, and the sepal_width range is set by those same 4 points. Discretising petal_length was rejected because it converts a continuous measurement into categories and loses information: 8 flowers (5.4%) would be placed in a bin dominated by another species. Each selected step supports the RQ by giving the models cleaner and more comparable inputs.

2.2. Correlation Analysis

Our variable set contains the four measured attributes plus the target, *species*: 5 variables, 10 unique pairs and therefore 40 values. The four measurements are the only candidate predictors available, so all are relevant to the RQ. petal_area is excluded from the correlation set because it is derived from two variables already in it. Species is nominal, so Pearson and Spearman are **not appropriate** for the four pairs that involve species: both require ordered (Pearson additionally interval-scaled) values, and any numeric coding of species would be arbitrary. For MI and NMI, each continuous variable was discretised into 5 equal-frequency bins (about 30 flowers per bin); MI is reported in bits (computed with natural logarithms and divided by ln 2) and NMI uses arithmetic-mean normalisation.

2.3. Supervised Learning Models and Evaluation

Target and features. The target is species (3 classes). The five features are the four measurements plus petal_area. After deduplication the classes are 50/50/49.

Train/test split. We used a stratified 80/20 split (random_state = 42), giving 119 training rows (40 setosa, 40 versicolor, 39 virginica) and 30 test rows (exactly 10 per class). Although the classes are almost balanced (largest-to-smallest ratio 50:49), with only 149 rows an unstratified split could give uneven class counts in a 30-row test set, so stratification keeps every class equally represented. A time-based split is not applicable because the data has no time attribute.

Validation strategy. Hyperparameters were tuned with 5-fold stratified cross-validation on the 119 training rows, scored by macro-F1. Each validation fold holds about 24 rows (roughly 8 per class). We did not use 10-fold CV because each fold would contain only about 12 rows (4 per class), and leave-one-out CV produces high-variance estimates on a dataset this small. For KNN, the scaler sits inside a pipeline so it is refitted on each training fold.

Hyperparameter grids. KNN: n_neighbors ∈ {1, 3, 5, 7, 9, 11, 15, 21, 31} × weights ∈ {uniform, distance} (18 configurations); k controls smoothness of the decision boundary, and weights controls how much close neighbours dominate the vote. Decision Tree: max_depth ∈ {1–6, None} × min_samples_leaf ∈ {1, 5} (14 configurations); both limit how finely the tree can partition the training data. The criterion was kept at its default (gini). The configuration with the highest mean CV macro-F1 was selected, with ties broken in favour of the simpler model.

Metrics, baseline and uncertainty. Macro-F1 is our primary metric because all three species are equally important to the RQ and it weights each class equally; per-class precision, recall and F1 and overall accuracy are also reported. The baseline is a majority-class (0R) classifier; because the training classes are 40/40/39 it predicts the first tied majority label, setosa, for every flower. Uncertainty in test macro-F1 is estimated with a percentile bootstrap (1,000 resamples of the 30 test predictions, 95% interval). We chose the bootstrap because it requires no retraining and makes no normality assumption, whereas repeated CV would measure variation across training sets rather than in the reported test score. We did not add an optional third model because the two required models already reach high performance.

2.4. Feature Selection

Embedded method: impurity-based feature importances from the Decision Tree. The tuned tree (max_depth = 2) splits on only two features and assigns 0.000 to the other three, so the full ranking

(including third place) was taken from an unpruned tree (max_depth = None) fitted to the same 119 training rows. **Filter method:** mutual information between each of the five features and species, estimated with scikit-learn's mutual_info_classif (nearest-neighbour estimator, random_state = 42, values in nats) on the training rows only. The top 3 features of each method are reported in Table 7.

2.5. Dimensionality Reduction & Clustering

We compared three candidate feature sets (Table 9): (A) all four measurements, (B) petal-only (petal_length, petal_width, petal_area) and (C) sepal-only (sepal_length, sepal_width). All sets were z-score standardised and evaluated with K-Means at k = 3 by counting how many flowers belong to the majority species of their cluster. We selected **Set A** because it uses all four measured attributes and therefore gives the most complete description of flower shape, and avoids hand-picking only the features we already expect to separate species.

k was chosen with the elbow method over k = 1–8 (Figure 1). Hierarchical clustering used agglomerative complete linkage with Euclidean distance and the same k = 3 chosen by the K-Means elbow method. PCA was applied to the same standardised Set A, and all four components were computed so that explained variance could be reported for the first three.

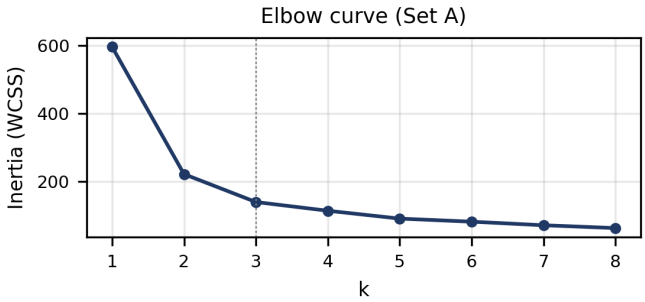


Figure 1: Elbow curve of K-Means inertia for k = 1–8, feature Set A (standardised). Dotted line marks the chosen k = 3.

3. Results Exploration and Analysis

3.1. Data Preprocessing

Table 2 shows the measurable effect of each implemented step.

Table 2: Before/after impact of the three implemented preprocessing steps.

Step	Before	After
Duplicate removal	150 rows; virginica 50	149 rows; virginica 49
petal_area engineered	Highest single-feature MI with species: petal_width 1.201 bits (NMI 0.619)	petal_area 1.238 bits (NMI 0.634)
Standardisation	Feature SDs 0.436–1.768; KNN (k = 5) CV macro-F1 = 0.955	All SDs = 1.000; KNN (k = 5) CV macro-F1 = 0.946

3.2. Correlation Analysis

Table 3 reports all four methods for the 10 pairs.

Table 3: Pearson, Spearman, MI (bits) and NMI for every pair of the five variables (n = 149). N/A = method not appropriate (nominal target).

Variable pair	Pearson	Spearman	MI (bits)	NMI
sepal_length – sepal_width	-0.118	-0.168	0.278	0.124
sepal_length – petal_length	0.874	0.884	0.843	0.365
sepal_length – petal_width	0.821	0.837	0.774	0.336
sepal_width – petal_length	-0.426	-0.309	0.358	0.159
sepal_width – petal_width	-0.363	-0.287	0.449	0.200
petal_length – petal_width	0.963	0.938	1.218	0.529
sepal_length – species	N/A	N/A	0.625	0.320
sepal_width – species	N/A	N/A	0.363	0.193
petal_length – species	N/A	N/A	1.196	0.615
petal_width – species	N/A	N/A	1.201	0.619

3.3. Supervised Learning Models and Evaluation

Table 4 lists the mean 5-fold CV macro-F1 for every configuration tried; Figure 2 plots training and validation scores for the uniform-weight KNN and the min_samples_leaf = 1 tree.

Table 4: Mean 5-fold stratified CV macro-F1 for every hyperparameter value tried (training-fold scores in brackets). Selected configurations shaded.

KNN (weights)			Decision Tree (min_samples_leaf)		
k	uniform	distance	max_depth	leaf = 1	leaf = 5
1	0.929 (1.000)	0.929	1	0.557 (0.557)	0.557
3	0.964 (0.966)	0.964	2	0.946 (0.983)	0.946
5	0.946 (0.966)	0.946	3	0.935 (0.983)	0.946
7	0.955 (0.962)	0.955	4	0.929 (0.996)	0.946
9	0.946 (0.958)	0.946	5	0.938 (1.000)	0.946
11	0.946 (0.962)	0.946	6	0.938 (1.000)	0.946
15	0.938 (0.962)	0.946	None	0.938 (1.000)	0.946
21	0.946 (0.962)	0.946			
31	0.929 (0.929)	0.954			

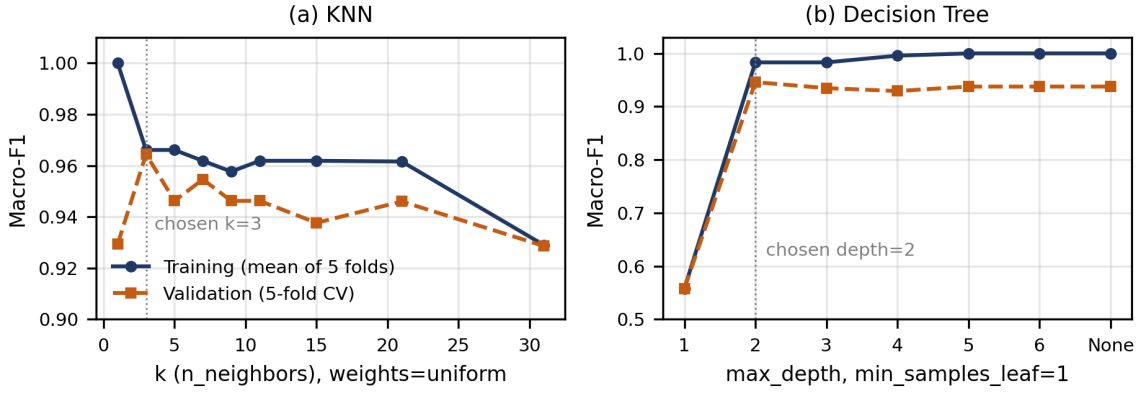


Figure 2: Training vs validation macro-F1 across hyperparameter values. (a) KNN, weights = uniform; (b) Decision Tree, min_samples_leaf = 1.

Table 5 states every hyperparameter changed from its scikit-learn default and its effect on CV macro-F1.

Table 5: Hyperparameters changed from default, and effect on mean 5-fold CV macro-F1.

Model	Hyperparameter	Default	Chosen	CV macro-F1 (default → chosen)
KNN	n_neighbors	5	3	0.946 → 0.964 (+0.018)
KNN	weights	uniform	uniform (kept)	—
Decision Tree	max_depth	None	2	0.938 → 0.946 (+0.008)
Decision Tree	min_samples_leaf	1	1 (kept)	—

Table 6 reports test-set performance. KNN misclassified one flower (id 135, virginica predicted as versicolor); the Decision Tree misclassified two (ids 57 and 78, versicolor predicted as virginica).

Table 6: Test-set (n = 30) per-class precision, recall and F1, with accuracy and macro-F1 95% bootstrap CI.

Model	Class	Precision	Recall	F1	Accuracy	Macro-F1 95% CI
0R baseline	setosa	0.333	1.000	0.500		
	versicolor	0.000	0.000	0.000		
	virginica	0.000	0.000	0.000		
	macro avg	**0.111**	**0.333**	**0.167**	0.333	—
KNN (k = 3)	setosa	1.000	1.000	1.000		
	versicolor	0.909	1.000	0.952		
	virginica	1.000	0.900	0.947		
	macro avg	**0.970**	**0.967**	**0.967**	0.967	[0.878, 1.000]
Decision Tree (depth 2)	setosa	1.000	1.000	1.000		
	versicolor	1.000	0.800	0.889		
	virginica	0.833	1.000	0.909		
	macro avg	**0.944**	**0.933**	**0.933**	0.933	[0.823, 1.000]

3.4. Feature Selection

Table 7: Top 3 features from the embedded (Decision Tree importance) and filter (mutual information, nats) methods.

Rank	Embedded: feature	Tuned tree (depth 2)	Unpruned tree	Filter: feature	MI (nats)
1	petal_width	0.527	0.502	petal_area	1.023
2	petal_area	0.473	0.481	petal_width	1.001
3	sepal_length	0.000	0.017	petal_length	1.000

The hard case we identified is test flower id 135 (Table 8).

Table 8: Attribute values and predictions for the hard-case flower (id 135).

id	sepal_length	sepal_width	petal_length	petal_width	petal_area	True	KNN	DT
135	6.1	2.6	5.6	1.4	7.84	virginica	versicolor	virginica

3.5. Dimensionality Reduction & Clustering

Table 9: Candidate clustering feature sets (standardised, K-Means k = 3).

Set	Features	Flowers in cluster's majority species
A (selected)	sepal_length, sepal_width, petal_length, petal_width	125 / 149 (83.9%)
B	petal_length, petal_width, petal_area	144 / 149 (96.6%)
C	sepal_length, sepal_width	116 / 149 (77.9%)

Inertia fell from 596.0 (k = 1) to 221.0 (k = 2), 138.8 (k = 3) and 113.1 (k = 4) (Figure 1). Table 10 profiles the resulting clusters and Figure 3 shows them in PCA space.

Table 10: Cluster sizes, mean measurements (cm) and species composition for K-Means and Hierarchical (complete linkage), k = 3.

Method	Cluster	Size	sepal_length	sepal_width	petal_length	petal_width	setosa / versicolor / virginica
K-Means	1	50	5.01	3.43	1.46	0.25	50 / 0 / 0
	0	52	5.80	2.67	4.36	1.40	0 / 39 / 13
	2	47	6.78	3.10	5.51	1.97	0 / 11 / 36
Hierarchical	1	49	5.02	3.45	1.47	0.24	49 / 0 / 0
	2	24	5.51	2.47	3.87	1.17	1 / 21 / 2
	0	76	6.48	2.99	5.18	1.81	0 / 29 / 47

Table 11: PCA on standardised Set A: explained variance ratio (EVR) and loadings. Top 3 loadings by magnitude in bold.

Component	EVR	Cumulative	sepal_length	sepal_width	petal_length	petal_width
PC1	0.730	0.730	**0.522**	-0.268	**0.580**	**0.565**
PC2	0.229	0.958	**0.373**	**0.925**	0.024	**0.068**
PC3	0.036	0.995	**0.720**	**-0.240**	-0.141	**-0.635**

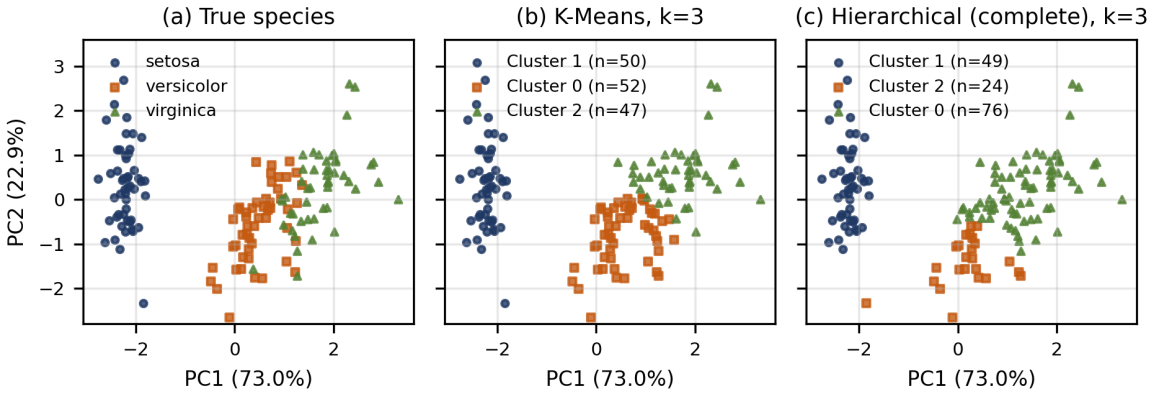


Figure 3: Flowers projected onto PC1 and PC2, coloured by (a) true species, (b) K-Means cluster and (c) Hierarchical cluster. Cluster colours are matched to the majority species.

4. Discussion and Interpretation

4.1. Data Preprocessing

Removing the duplicate reduced the dataset from 150 to 149 rows, which ensures every flower is counted once and that cross-validation is not biased by repeated observations. Engineering petal_area increased the highest single-feature MI with species from 1.201 to 1.238 bits, showing that combining petal measurements creates a more informative feature for predicting species. Standardisation put all features on the same scale (SD = 1.000), so that each measurement contributes equally to KNN distances; KNN CV macro-F1 changed from 0.955 to 0.946, which is a very small difference. Overall, the preprocessing produced a cleaner and more consistent dataset for answering the RQ.

4.2. Correlation Analysis

Agreement and divergence. Pearson and Spearman agree closely for most pairs, for example petal_length–petal_width (0.963 vs 0.938) and sepal_length–petal_length (0.874 vs 0.884). This is expected because Pearson measures linear relationships while Spearman measures monotonic relationships, and the strong relationships in this data are approximately linear. The largest divergence is sepal_width–petal_length (-0.426 vs -0.309), which may be due to outliers affecting Pearson more than Spearman. MI and NMI rank the pairs similarly to the correlation coefficients, with petal_length–petal_width having the highest MI (1.218 bits). MI can also detect non-linear relationships, which is why it is useful alongside Pearson and Spearman.

Connection to the RQ. The strongest associations with species are for petal_width (NMI 0.619) and petal_length (NMI 0.615), while sepal_width has the weakest (NMI 0.193). This suggests that petal measurements are more strongly associated with species than sepal measurements. sepal_length and sepal_width are almost uncorrelated (Pearson -0.118), indicating that sepal length and width vary largely independently. These are associations only; other factors such as growing conditions or plant age, which are not in the dataset, may confound these relationships.

Impact on next steps. Because petal_length and petal_width are highly correlated (r = 0.963), they carry overlapping information, which motivated combining them into petal_area and prioritising petal features in feature selection. We kept sepal_width in the models despite its weak association with species (NMI 0.193), because it may still contribute in combination with other features.

4.3. Supervised Learning Models and Evaluation

Performance and baseline. Both models far exceed the 0R baseline. KNN reached a test macro-F1 of 0.967 and the Decision Tree 0.933, compared with 0.167 for 0R: absolute improvements of 0.800 and 0.766 respectively. In accuracy terms, KNN improved on 0R by 0.634 (0.967 vs 0.333) and the tree by 0.600. Setosa was classified perfectly by both models (F1 = 1.000), while all errors involved versicolor and virginica, showing these two species are harder to separate.

Bias and variance. Figure 2 shows the expected trade-off. KNN with k = 1 achieves a training score of 1.000 but a CV score of only 0.929, indicating overfitting (high variance), while k = 31 has a training score of 0.929, indicating underfitting (high bias). The Decision Tree with max_depth = 1 scores 0.557 on both training and validation folds, which is clear underfitting, while unlimited depth reaches 1.000 on training but 0.938 on validation. Our chosen values (k = 3, depth = 2) balance bias and variance.

Uncertainty. The bootstrap 95% CI for KNN macro-F1 is [0.878, 1.000] and for the tree [0.823, 1.000]. Both lower bounds are well above the baseline of 0.167, which shows that both models perform reliably and that KNN is the better model.

Feature influence, behaviour and limitations. The Decision Tree relies on petal_width (importance 0.527) and petal_area (0.473), while KNN uses all five features equally in its distance calculation. The tree is highly interpretable: its first split (petal_width ≤ 0.80) isolates setosa exactly. KNN is less interpretable and is sensitive to noise and irrelevant features, but it achieved higher macro-F1 (0.967 vs 0.933). Decision Trees can overfit, which we controlled by limiting depth to 2. Overall, the results answer the RQ positively: species can be predicted from flower measurements with high accuracy, and petal measurements are the most important.

4.4. Feature Selection

Comparing the lists. Both methods agree that petal_width and petal_area are among the top features, but they disagree on third place: the embedded method ranks sepal_length third (0.017) while the filter method ranks petal_length third (MI 1.000). This happens because the two methods measure importance differently: the filter method scores each feature on its own relationship with species, whereas the embedded method measures how much each feature is used by the trained model.

Hard case. Flower id 135 is a virginica with petal_width = 1.4 and petal_area = 7.84. Under the embedded method's top feature (petal_width), a value of 1.4 is more typical of versicolor, whereas under the filter method's top feature (petal_area), 7.84 lies above the tree's split point of 7.42 and points to virginica. KNN predicted versicolor and the tree predicted virginica. This row is hard because its measurements lie near the boundary between versicolor and virginica, where the two species overlap.

Opposite rankings. A filter method could rank a feature highly while the embedded method ranks it low when the feature is strongly related to the target but correlated with another feature. In that case, the model may use only one of the correlated features and give the other a low importance. In our feature set, this could apply to petal features, which are highly correlated (r = 0.963).

4.5. Dimensionality Reduction & Clustering

PCA. PC1 explains 73.0% of the variance and has its largest loadings on petal_length (0.580), petal_width (0.565) and sepal_length (0.522), so it represents an overall "flower size" axis. PC2 explains 22.9% and is dominated by sepal_width (0.925), followed by sepal_length (0.373) and petal_width (0.068), so it represents a "sepal width" axis. PC3 explains 3.6% with loadings on sepal_length (0.720), petal_width (-0.635) and sepal_width (-0.240), contrasting sepal length with petal width. Together the first two components explain 95.8% of the variance. Figure 3(a) shows that species separate mainly along PC1, which supports the finding that size-related petal measurements are most relevant to the RQ.

Clusters. Using the dominant features in Table 10, K-Means cluster 1 (n = 50) contains small-petalled flowers (mean petal_length 1.46) and matches setosa exactly. Cluster 0 (n = 52) has medium petals (4.36) and is mostly versicolor (39 of 52), while cluster 2 (n = 47) has the largest petals (5.51) and is mostly virginica (36 of 47). The clusters therefore represent small, medium and large flowers. Hierarchical clustering produced a setosa cluster of 49, a medium cluster of 24 and a large cluster of 76. Both methods produced broadly similar groupings, with a clear setosa cluster and two overlapping clusters for versicolor and virginica, which is consistent with the classification results.

Relevance to the RQ. The clusters show that natural groupings in the measurements partly correspond to species (125 of 149 flowers, 83.9%, belong to their K-Means cluster's majority species). This supports the RQ by showing that species differ in flower size, but the overlap between clusters 0 and 2 shows that unsupervised methods cannot fully recover the species labels.

5. Limitations and Improvement Opportunities

5.1. *Preprocessing*

Removing duplicates may remove genuine observations if two flowers happened to have identical measurements. petal_area is a simplification, as petals are not rectangles, so the feature may not accurately represent true petal area. Standardisation assumes features are approximately normally distributed. In future, other feature engineering approaches could be explored.

5.2. *Correlation*

The dataset is small (149 rows), so correlation estimates may not generalise. Pearson correlation is sensitive to outliers and only measures linear relationships. MI and NMI depend on the discretisation method and the number of bins chosen. Correlation does not imply causation, and confounding variables may exist.

5.3. *Supervised Learning*

The test set of 30 rows is small, so results may vary with a different random split. The models could overfit the training data, and only two models were compared. Other models such as SVM or Random Forest might perform better. Additional features, such as colour or location, are not available in the dataset and might improve performance.

5.4. *Feature Selection*

Decision Tree importances can be biased and may be unstable when features are correlated. Mutual information estimates depend on the estimator and random seed. Only two feature selection methods were used; wrapper methods such as recursive feature elimination could provide a different ranking.

5.5. *PCA & Clustering*

PCA only captures linear relationships and the first two components do not explain all of the variance. Results are sensitive to scaling. K-Means assumes spherical clusters of similar size, and the choice of k affects the results. Other clustering methods such as DBSCAN could be explored in future work.

6. Conclusion

This report investigated whether iris species can be predicted from flower measurements and which measurements distinguish species most strongly. Correlation analysis showed that petal_width (NMI 0.619) and petal_length (NMI 0.615) are most strongly associated with species. Both models clearly outperformed the 0R baseline (macro-F1 0.167): KNN achieved a macro-F1 of 0.967 and the Decision Tree 0.946, so KNN is the better model for this task. Feature selection identified petal_width and petal_area as the most important features under both methods, and PCA and clustering showed that species differ mainly in flower size. Overall, the answer to our RQ is yes: species can be predicted accurately, and petal measurements are the most important attributes.

7. References

Fisher, R. A. (1936). The use of multiple measurements in taxonomic problems. *Annals of Eugenics, 7*(2), 179–188. https://doi.org/10.1111/j.1469-1809.1936.tb02137.x

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., … Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825–2830.